Introduction / Background

CSCI 4022 Project Report
Group: Brooke Seifert, Katie Miller, and Brooke Nelson

For our CSCI 4022 Data Science Project, we chose to explore Spotify data to understand how songs can be grouped by audio characteristics and patterns in listening behavior. Working with song data and answering questions about music taste is interesting to us because we are frequent listeners to music on Spotify, and we wanted to use our data science skills to analyze song features and learn more about the relationships among the music that people listen to.

The methods we used for this project are K-means clustering and Market Basket Analysis. K-means clustering is a method that groups similar data points into clusters without needing labeled data. It is used to uncover hidden patterns when the goal is to organize data based on similarity. Market Basket Analysis is a technique used to identify products frequently purchased together. In this scenario, we treat playlists as transactions and tracks as products.

The dataset we used was “30000 Spotify Songs” on Kaggle from the Spotify API. This dataset contains approximately 30,000 songs along with audio features such as danceability, energy, and tempo. The data was obtained from Spotify and then uploaded to the Kaggle website. This dataset is observational and static, meaning that it was recorded without researcher intervention or manipulation, and the data will not change since it was uploaded.

We aim to answer multiple questions during our exploration of this dataset. We are curious how the “vibe” of a playlist is defined by certain features of its songs, and whether similar songs tend to appear in groups or are typically grouped. We used K-means and Market Basket Analysis to answer questions concerning how playlists vary by “vibe” and how to relate songs. Specifically, we used K-means to represent playlists by their song features and cluster by music taste, analyzing how many different clusters we typically get across playlists, along with comparing and determining relationships between multiple attributes. Market Basket Analysis was used to find song combinations that frequently occur together in playlists and to generate association rules for music recommendations.

Prior work on music recommendation systems or on Spotify data analysis helped us in our process. This article discusses playlist relationships and explains code snippets that utilize song features. Additionally, this analysis concerns relationships between popularity across different platforms and provides helpful initial visualizations.

Methods and Results

To prepare this data, we downloaded it from Kaggle and then cleaned the dataset. We loaded the CSV file into our Jupyter notebook and converted it into a dataframe. After this, we altered the dataset by removing duplicates and null values. We then grouped the dataframe by playlist_id and created a dictionary that maps each playlist to its list of songs.

As mentioned, the methods that are appropriate for the questions we aim to answer are K-means clustering and Market Basket Analysis. For K-means clustering, we assumed that the similarity between songs could be efficiently measured using quantitative audio characteristics, such as danceability and tempo. We chose pairs of features that we believed could be compared on a similar scale without one dominating over the other (for example, danceability and energy). Since K-means also assumes fairly spherical clusters with roughly even spread, we chose to test multiple configurations rather than just one to see whether insightful groups appeared. For Market Basket Analysis, we assumed that playlists could be treated as transactions and that songs that appeared together in playlists indicated meaningful relationships. Since playlists typically consist of songs with similar "vibes", this assumption was reasonable.

K-Means

Tempo and Danceability vs Tempo and Energy

According to the documentation in the Kaggle dataset, danceability "describes how suitable a track is for dancing based on a combination of musical elements, including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable, and 1.0 is most danceable." Similarly, energy "is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy." Finally, tempo describes a track "in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration."

Computing K-Means For Each Song Based on Two Features

The question we aimed to answer with the first iteration of K-means was whether a certain song is usually grouped in the same cluster for different features. To answer this, we performed K-means on each song, clustering on danceability and tempo, and energy and tempo. After using the Elbow Method to determine the best k-value by choosing where the slope of the WCSS values begins to flatten out, the songs were clustered separately.

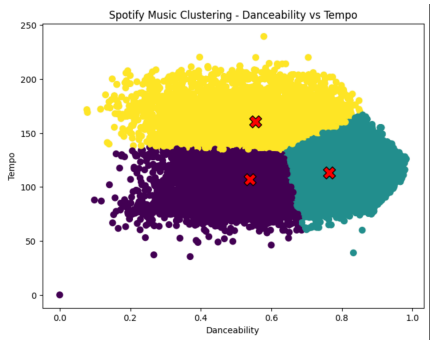

Figure 1: Song Clustering Danceability vs Tempo

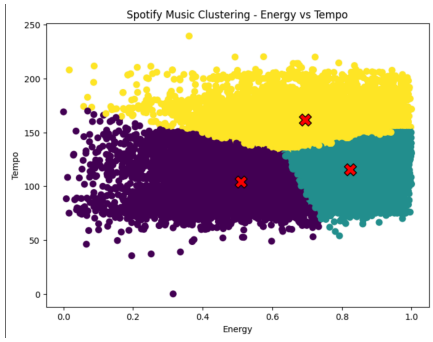

Figure 2: Song clustering Energy vs Tempo

After saving cluster assignments for every song for both iterations, we computed the average energy and danceability to determine cluster rankings. The cluster rankings show which clusters correspond to high, medium, and low values of energy or danceability. This was determined by computing the average danceability and energy for each cluster group. We determined that the clusters were ordered from highest average to lowest average as Cluster 1, Cluster 2, and finally Cluster 0 for both features. This means that the songs with high energy and high danceability were both put in their respective Cluster 1's. After grouping and determining the rankings of the clusters, we calculated how many songs had matching cluster assignments for energy and danceability. After calculations, 55.24% of all songs had matching assignments. In about 55% of the tracks in this dataset, energy and danceability move together as one. A track with both classifications in cluster 1 is songs with high energy and danceability. A track with both classifications in cluster 0 has low energy and low danceability. The tracks that are outside the 55% are the songs that are not correlated in the two features. These are the songs that do not follow a positive correlation between energy and danceability. A few features, such as energy and danceability, are not enough to define the mood or "vibe" of a song. Since there is about a 45% discrepancy in songs that do not match feature rankings, we can argue that one cannot reliably predict the other. Since almost half the songs are not in the same cluster categorization for energy and danceability, we conclude that a song cannot be defined by just two features.

Computing K-Means Per Playlist Based on Two Features

After concluding that a song cannot be defined by only two of its features, we moved into exploring how playlists are defined by different "vibes" using K-means with a focus on the variability of the number of k-values. Our driving questions concerned how many different "vibes" exist in a playlist in terms of tempo and danceability/energy, and how the number of clusters changes based on the parameters chosen. We addressed these questions by exploring differing k-values for each playlist and establishing an optimal value. With the optimal k-value per playlist, we found the average across all playlists when clustering based on tempo and danceability, and compared it to the average based on tempo and energy.

To find the ideal k-value, we calculated the silhouette score. We chose to use silhouette score instead of a method like the Elbow Method for this section since it concerned 471 playlists. With such a large number of playlists, visualizing the graph for the Elbow Method would be unreasonable and less consistent across playlists since it is dependent on visual estimations. Instead, the method used for finding the optimal k-value per playlist we used was calculating silhouette score and taking the k-value that resulted in the highest score. This provided consistency across playlists and allowed for faster computation.

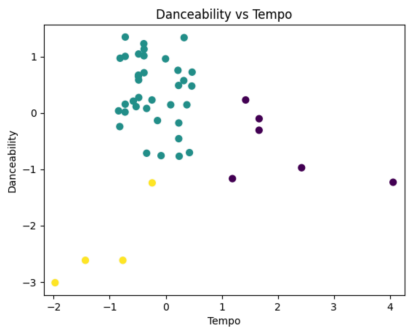

Figure 3: Playlist #3 Danceability vs Tempo

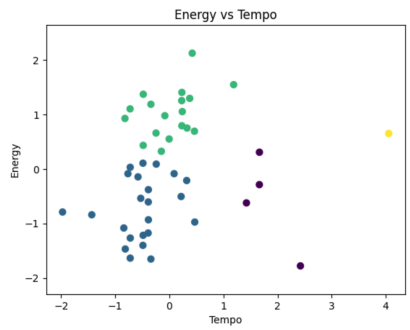

Figure 4: Playlist #3 Energy vs Tempo

With the ideal k-value estimated for each playlist, we could then compare clustering based on danceability vs tempo as opposed to clustering on energy vs tempo. As seen in Figure 3 and Figure 4, there are not only different clusters but also different numbers of clusters, which is what we were aiming to consider when asking about the number of "vibes" that exist in a playlist. Differences in the number of clusters between the two chosen parameters were consistent across playlists, but the number of clusters only seemed to vary by a degree of about one. Finally, clustering on danceability and tempo across each playlist yielded an average number of clusters of about 3.71. Similarly, clustering on energy and tempo yielded an average number of clusters of approximately 4.17. Ultimately, this is a very similar number of clusters and therefore a similar number of "vibes" according to danceability, energy, and tempo per playlist.

Concerning our initial questions about how many different "vibes" exist in a playlist, we were able to determine that, based on tempo, danceability, and energy, there typically are around 4 different "vibes" in a playlist. However, there is negligible difference between clustering on danceability and energy, which informs us that they cover a similar structure in the data. This also suggests that tempo likely plays a more significant role in establishing the "vibe" of a playlist since it was relatively consistent across both comparisons. Overall, it is evident that slight variations in features do not greatly affect the total number of clusters in a playlist, and instead, there are reliably a relatively low number of "vibes" in playlists.

Additionally, we compared the k-value to the number of songs in a playlist to determine if there was a clear positive relationship between the number of songs in a playlist and the best k-value for that playlist. Visualization through a scatter plot in Figure 5 shows that there is no clear relationship between song count and k-value.

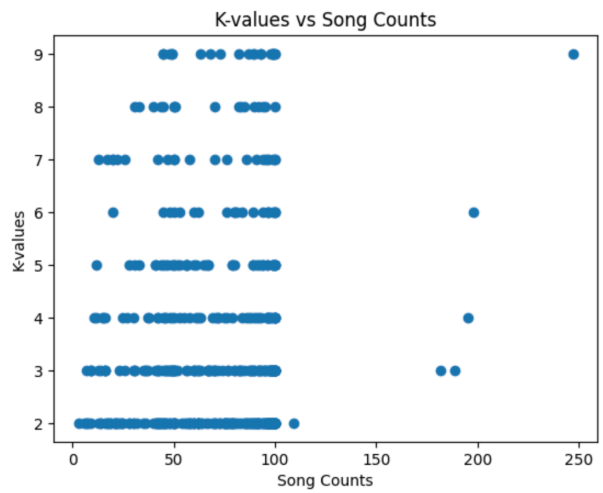

Figure 5: K-Values vs Song Counts

Market Basket Analysis

We used Market Basket Analysis to address our questions regarding listening patterns and music recommendations. To understand listening patterns, we first needed to see how and which songs grouped together within certain playlists. Then, we could use these groupings to find rules that define which songs appear together. This part of the process is what drives song recommendation based on the current songs in a playlist. To accomplish this, we used the Apriori algorithm to identify frequent itemsets and derive association rules from them. As mentioned, we treated playlists as transactions and songs as items. Therefore, for our purposes, an itemset is a group of songs that appear together in the same playlist. First, we observed the top frequent itemsets in our dataset.

```
=== Top 10 Frequent Itemsets (min_support = 0.01) ===
      support                      itemsets  length
50   0.044586          frozenset({Breathe})       1
304  0.044586           frozenset({Poison})       1
8    0.042463            frozenset({Alive})       1
145  0.042463          frozenset({Forever})       1
291  0.040340         frozenset({Paradise})       1
385  0.038217             frozenset({Stay})       1
12   0.033970            frozenset({Alone})       1
179  0.031847           frozenset({Heaven})       1
225  0.031847  frozenset({Let Me Love You})       1
237  0.031847             frozenset({Lost})       1
```

Figure 6: Top 10 Frequent Itemsets

Figure 6 shows the top 10 frequent itemsets, each an individual song. The support for these itemsets is around 3-4%. This implies that the dataset is large and diverse, and even individual songs don't appear that frequently. We found that a minimum support threshold of 0.01 was sufficient because it didn't filter out too many songs or song combinations.

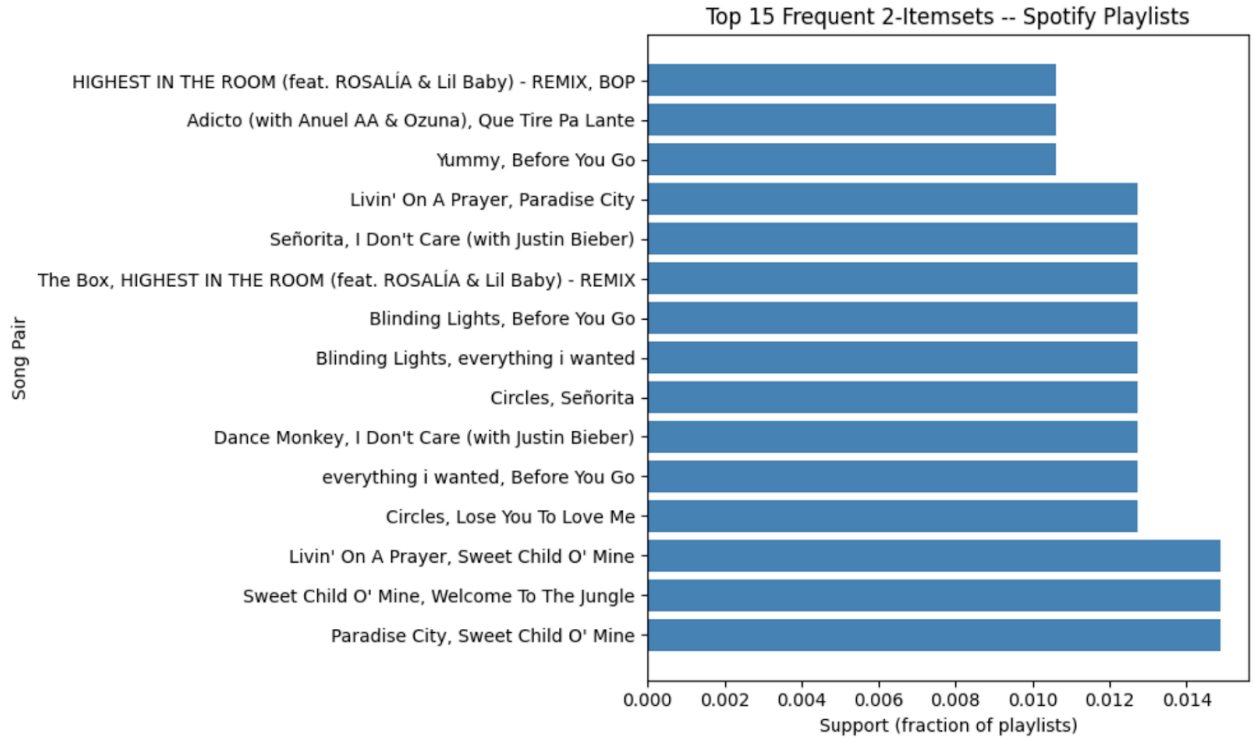

Figure 7: Support of Top 15 Frequent 2-Itemsets

This observation is further supported by the top 15 frequent 2-itemsets visualized in Figure 7, which have a support of around 1-1.4%. Acknowledging that the structure of the dataset we chose impacts our support is important for interpreting our association rules. Overall, we found 545 frequent itemsets across itemset lengths 1 to 4 using a minimum support of 5.

```
=== Top 10 Association Rules (confidence >= 0.5) ===
                                                antecedents                                                consequents  support  confidence  lift
                     frozenset({Adicto (with Anuel AA & Ozuna)})                                 frozenset({Que Tire Pa Lante}) 0.010616    1.000000  94.2
                                 frozenset({Que Tire Pa Lante})                       frozenset({Adicto (with Anuel AA & Ozuna)}) 0.010616    1.000000  94.2
                            frozenset({Señorita, Dance Monkey})     frozenset({Don't Start Now, I Don't Care (with Justin Bieber)}) 0.010616    1.000000  94.2
    frozenset({Don't Start Now, I Don't Care (with Justin Bieber)})                                 frozenset({Señorita, Dance Monkey}) 0.010616    1.000000  94.2
                       frozenset({Life Is Good (feat. Drake)}) frozenset({HIGHEST IN THE ROOM (feat. ROSALÍA & Lil Baby) - REMIX}) 0.010616    1.000000  78.5
frozenset({HIGHEST IN THE ROOM (feat. ROSALÍA & Lil Baby) - REMIX})                       frozenset({Life Is Good (feat. Drake)}) 0.010616    0.833333  78.5
frozenset({HIGHEST IN THE ROOM (feat. ROSALÍA & Lil Baby) - REMIX})                                           frozenset({The Box}) 0.012739    1.000000  78.5
                                         frozenset({LA CANCIÓN})                                 frozenset({Soltera - Remix}) 0.010616    0.833333  78.5
                                             frozenset({The Box})                       frozenset({Life Is Good (feat. Drake)}) 0.010616    0.833333  78.5
                                   frozenset({Before You Go})           frozenset({everything i wanted, Blinding Lights, Yummy}) 0.010616    0.833333  78.5
```

Figure 8: Top 10 Association Rules

As shown in Figure 8, the top 10 association rules had very high confidence and lift but low support. The highest support for an association rule is around 1.1%, which reflects our observation that the dataset we chose is large and diverse. Overall, these rules are rare but strong. When they do occur, they happen more often than would be expected by random chance. For example, if a user has "Adicto (with Anuel AA & Ozuna)" in their playlist, a good

recommendation would be “Que Tire Pa Lante” because “Que Tire Pa Lante” appeared in a playlist every time “Adicto (with Anuel AA & Ozuna)” appeared. The results of the Apriori algorithm show us that there is a lot of variety in listening behavior, and that certain songs are very likely to be paired together. Based on our dataset, listening patterns and song groupings do exist, and this implies that effective music recommendations can be generated.

Conclusions

In our exploration of our dataset, we made a few conclusions after our analysis. In response to questions about how the “vibe” of a playlist is defined by features, and wondering whether similar songs are typically grouped, we concluded that a song cannot be defined by just two features, and based on tempo, danceability, and energy, there are typically around 4 different “vibes” in a playlist. Regarding our questions on which songs frequently occur together in playlists and whether we can use these patterns to make music recommendations, we determined that certain songs consistently occur together in playlists, and these patterns can be used to generate music recommendations.

Something that surprised us was the fact that energy and danceability are not very commonly correlated in intensity. As previously mentioned, only about 55% of songs had matching danceability and energy intensity. However, when considered through clustering on individual playlists, the overall “vibe” of a playlist could be consistently separated into approximately 4 groups based on both tempo and danceability as well as tempo and energy. While danceability and energy do not move together overall, they do lead to similar distinctions in playlists. Through this project, we were able to gain insight into how music and listening behavior can be analyzed using data science methods like K-means clustering and Market Basket Analysis. We were able to answer our main research questions based on our data, but given more time, we would like to incorporate more features and look at additional datasets. We could potentially expand on this project by utilizing Gaussian Mixture Models instead of K-means to avoid circular clusters and soft clustering. We could also try clustering on different features to see if we have more variability and correlation between song attributes.

Work Distributions

For this project, the work was distributed as follows. The K-means clustering on every song comparing danceability and tempo and energy and tempo was completed by Katie Miller. The K-means clustering per playlist on tempo and danceability and tempo and energy was completed by Brooke Seifert. All of the code relating to Market Basket Analysis was written by Brooke Nelson. Our presentation was outlined by Brooke Nelson, our completed code notebook was formatted by Brooke Seifert, and the outline of the report along with the introduction and conclusions were completed by Katie Miller.